

ALEKSANDROVSKIY ZAVOD, Russian Federation

WMO: 309710

Lat: 50.917N

Lon: 117.933E

Elev: 2646

StdP: 13.34

Time Zone: 9.00 (E09)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.5	-32.9	-42.7	0.5	-37.1	-37.9	0.7	-32.7	17.8	3.3	15.5	-2.0	2.3	200	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.2	83.4	63.3	79.9	62.5	76.7	61.8	67.1	77.5	65.4	75.3	63.8	73.1	7.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	97.6	72.0	62.0	91.6	69.6	60.3	86.1	67.9	33.3	77.5	31.9	75.4	30.6	73.2	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	16.6	14.4	DB	-42.2	89.4	4.1	4.3	-45.2	92.5	-47.6	95.0	-50.0	97.4	-53.0	100.5	(4)
(5)			WB	-42.4	70.6	4.1	1.4	-45.3	71.5	-47.7	72.3	-50.0	73.1	-53.0	74.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	-13.0	-6.9	10.2	31.0	46.3	58.2	62.9	58.6	46.3	28.3	5.0	-10.2	(6)	
	DBStd	28.63	8.17	9.44	12.23	9.14	7.67	6.59	5.02	6.37	7.79	9.70	11.53	8.76	(7)	
	HDD50	9585	1953	1594	1233	571	162	11	1	11	160	672	1350	1867	(8)	
	HDD65	14089	2418	2014	1698	1019	581	218	99	212	562	1136	1800	2332	(9)	
	CDD50	1035	0	0	0	2	49	257	399	278	49	1	0	0	(10)	
	CDD65	64	0	0	0	0	2	15	33	13	1	0	0	0	(11)	
	CDH74	1340	0	0	0	2	59	431	540	268	40	0	0	0	(12)	
	CDH80	312	0	0	0	0	9	122	122	53	6	0	0	0	(13)	
	WSAvg	5.1	3.3	3.9	5.6	7.2	7.6	5.5	4.8	4.9	5.5	5.6	4.4	3.3	(14)	
	PrecAvg	15.1	0.2	0.2	0.3	0.5	1.3	2.6	3.8	3.6	1.6	0.5	0.3	0.2	(15)	
(16) Precipitation	PrecMax	23.7	0.5	0.6	0.7	1.5	3.6	8.3	7.6	8.2	5.4	1.7	1.0	0.6	(16)	
	PrecMin	9.7	0.0	0.0	0.1	0.0	0.0	0.7	0.5	1.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	3.4	0.1	0.1	0.2	0.4	0.9	1.6	1.6	1.7	1.3	0.4	0.2	0.2	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	24.1	50.4	69.0	81.0	90.0	88.3	85.5	79.6	63.0	40.7	21.5	(19)	
		MCWB	14.8	20.1	36.7	48.8	57.4	63.1	66.2	65.5	58.0	47.1	32.8	18.6	(20)	
	2%	DB	9.0	18.9	42.3	60.9	74.3	82.8	83.2	80.4	72.8	54.9	33.5	12.0	(21)	
		MCWB	7.5	16.1	32.0	44.0	54.1	62.1	64.9	63.2	55.2	41.7	28.2	10.2	(22)	
	5%	DB	5.0	14.4	36.8	54.7	69.4	78.7	79.7	76.6	67.4	49.7	27.8	7.7	(23)	
		MCWB	3.7	12.1	29.1	40.3	51.2	60.5	64.6	62.7	52.5	39.2	24.4	6.4	(24)	
	10%	DB	1.0	10.6	31.2	49.0	64.7	74.7	76.5	72.8	62.8	44.5	23.2	3.8	(25)	
		MCWB	0.0	8.8	25.6	37.1	48.6	59.0	63.7	61.1	50.7	36.4	20.7	2.7	(26)	
	0.4%	WB	14.8	20.2	37.3	48.7	59.1	67.2	69.9	68.6	59.9	47.0	33.5	18.8	(27)	
		MCDB	16.1	24.0	48.6	67.7	77.4	84.4	80.9	80.0	72.6	61.5	39.8	21.5	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	7.6	16.2	32.9	44.2	55.5	64.7	67.9	65.9	57.3	42.9	28.6	10.4	(29)	
		MCDB	8.9	18.9	41.6	60.0	71.0	77.3	77.8	75.6	67.3	52.0	32.9	11.8	(30)	
	5%	WB	3.7	12.3	29.4	40.8	52.4	62.7	66.3	64.1	55.0	40.0	24.4	6.5	(31)	
		MCDB	4.9	14.4	36.6	54.1	66.9	73.7	75.6	73.5	64.4	48.7	27.7	7.6	(32)	
	10%	WB	0.1	8.9	25.7	37.6	49.5	60.9	64.8	62.2	52.2	37.1	20.7	2.8	(33)	
		MCDB	1.0	10.6	30.9	48.1	62.7	71.1	73.7	70.3	61.5	43.8	23.1	3.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	24.2	28.0	28.7	25.7	27.4	27.0	24.2	25.1	26.8	24.8	23.8	23.1	(35)	
		MCDBR	26.5	29.0	31.0	36.5	37.0	35.8	31.7	32.5	35.3	31.3	26.4	25.1	(36)	
		MCWBR	25.0	26.5	22.2	21.9	19.6	17.3	16.3	17.0	19.1	19.9	21.1	23.4	(37)	
		MCDBR	26.4	28.6	30.1	34.3	32.8	28.7	24.1	25.4	28.8	29.1	25.8	25.0	(38)	
	5% WB	MCWBR	24.9	26.1	22.0	21.1	18.5	15.0	13.6	14.6	16.4	19.1	21.0	23.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.247	0.273	0.320	0.416	0.484	0.465	0.416	0.386	0.344	0.313	0.276	0.254	(40)	
	taud		2.219	2.176	2.114	2.004	1.885	2.053	2.265	2.323	2.378	2.362	2.265	2.175	(41)	
	Ebn at Noon		249	272	276	259	246	253	265	267	268	255	233	222	(42)	
	Edh at Noon		22	30	39	50	59	50	40	36	30	25	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		495	862	1353	1667	1867	1951	1794	1601	1259	857	503	372	(44)	
	RadStd		19	35	65	93	136	177	119	118	91	52	23	20	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	-3.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page